

Gabrielle Johnson's Inputs / Outputs — Integration Notes

(a) What the model receives — and from where

Input	Source	Notes
Vitals	Device stream — patient smartwatch	Nurse accepts or rejects each value before it enters the model
Chief complaint	Patient entry at the kiosk, or nurse entry for those taking the nurse route	
Age, sex	Patient/nurse entry depending on mode	Age also gates the model: under-18 suppresses the suggestion entirely
Arrival mode	Patient/nurse entry	
Indoor location	Device stream — BLE/UWB anchors	Used only to find the patient until admission
Repeat vitals during the wait	Device stream , continuous from the watch	

(b) What the model emits

At triage — to the nurse's handheld device only

Output	Form
Suggested ESI level	A number, 1–5, with its target time to-be-seen (ESI 2 → "10 minutes")
Reasoning	Up to three features supporting the suggestion and up to two arguing against it, as labelled bars
Data-quality flags	Amber highlight on implausible data values
Blind-spot warning	A fixed amber card to highlight biases in the model which can help with inferring the system's predictive result

During the wait

Output	Destination	Form
Ambient attention state	Patient watch	Amber screen with an icon, plus a distinct vibration pattern
Clinical alert	Nurse handheld device	Vibration distinguishable from routine message traffic, opening a card with the trend that fired it (e.g. HR 88→118), the patient's location, a named owner, and a countdown to escalation
Escalation	Charge nurse's handheld device	Fires if the alert is unacknowledged, or if a checked-in patient goes unreviewed past the ESI

Output	Destination	Form
		target (≤ 10 min for ESI 2, ≤ 30 min for ESI 3 etc)
Status	Kiosk, on the patient scanning their watch	Queue patient's position or range

(c) What the human does next

The triage nurse, at assessment

1. Scans the patient's watch QR to pair the record.
2. **Accepts or rejects each captured vital**
3. **Accepts or declines ESI level.** Declining requires selecting a reason, eg. "patient looks worse than the data" etc. The reason is logged for audit and retraining.

The nurse receiving a deterioration alert

1. Feels the vibration, reads the trend and location.
2. **Acknowledges**, which stops the escalation clock and names her as owner or **hands off** if she cannot break away, which reassigns the patient.
3. Pauses her current assessment if needed; the system preserves its state so she can resume without re-entering data.
4. Goes to the patient and **re-triages**
5. If re-triaged, the new level re-enters the queue and the paper record.

The charge nurse

Receives escalations for unacknowledged alerts and overdue unseen patients, and reallocates.

The patient

1. May scan at the kiosk at any point for a position update.

Known gaps at this stage

- The model was trained on nurse measured vitals; watch measured vitals are a different measurement distribution and require revalidation before any live use.

- Pulse oximetry overestimates SpO_2 in darker skin